

GL Towers Dashboard

State Machine Specification

****Version 1.0****

Table of Contents

1. [Purpose and Scope](#1-purpose-and-scope)
2. [Notation](#2-notation)
3. [System-Level State Machine](#3-system-level-state-machine)
4. [Startup and Initialization](#4-startup-and-initialization)
5. [Telemetry State Machine](#5-telemetry-state-machine)
6. [Sensor Validation State Machine](#6-sensor-validation-state-machine)
7. [Health Evaluation State Machine](#7-health-evaluation-state-machine)
8. [Alert Generation State Machine](#8-alert-generation-state-machine)
9. [Recommendation Generation State Machine](#9-recommendation-generation-state-machine)
10. [Dashboard Refresh State Machine](#10-dashboard-refresh-state-machine)
11. [Screen Navigation State Machine](#11-screen-navigation-state-machine)
12. [Touch Interaction State Machine](#12-touch-interaction-state-machine)
13. [LED Configuration State Machine](#13-led-configuration-state-machine)
14. [Irrigation Configuration State Machine](#14-irrigation-configuration-state-machine)
15. [Communication Failure and Recovery State Machine](#15-communication-failure-and-recovery-state-machine)
16. [Pump Manual Control State Machine](#16-pump-manual-control-state-machine)
17. [Tank Adjustment State Machine](#17-tank-adjustment-state-machine)
18. [Notification Center State Machine](#18-notification-center-state-machine)
19. [Future Integration Points](#19-future-integration-points)
20. [State Summary Tables](#20-state-summary-tables)

1. Purpose and Scope

This document specifies the complete operational behavior of the GL Towers Dashboard as a set of formal state machines. Each state machine describes the states, events, transitions, conditions (guards), and outputs for one functional subsystem.

The firmware implementation must reproduce the behavior described in every state machine exactly in order to achieve functional equivalence with the reference dashboard.

This document is complementary to the Functional Logic Specification (Document 1). Where Document 1 describes **what** computations are performed, this document describes **when** those computations execute and what state transitions they produce.

2. Notation

State diagrams use Mermaid stateDiagram-v2 syntax.

Symbol	Meaning	
-----	-----	
`[*]`	Entry point or terminal state	
`State --> State`	Unconditional transition	
`State --> State : event [guard] / output`	Transition with event, optional guard, optional output	
`state "Label" as X`	State alias	
Note annotations	Supplementary explanation	

Guard conditions are written in square brackets. Outputs (actions) follow the slash. Multiple guards and outputs on the same transition are separated by newlines in the body text.

3. System-Level State Machine

The top-level state machine describes the major operational phases of the dashboard system.

```
``mermaid
```

```
stateDiagram-v2
```

```
    [*] --> Startup
```

```
    Startup --> Initializing : power on / load stored state
```

```
    Initializing --> WaitingForTelemetry : initialization complete
```

```
    Initializing --> Initializing : stored state loading
```

```
    WaitingForTelemetry --> Operating : first telemetry snapshot received
```

```
    WaitingForTelemetry --> WaitingForTelemetry : tick / no snapshot yet
```

```
    Operating --> Operating : telemetry tick / applySnapshot
```

```
    Operating --> CommunicationWarning : lastUpdate age > 120,000 ms
```

```
    Operating --> Shutdown : power off event
```

```
    CommunicationWarning --> Operating : telemetry snapshot received
```

```
    CommunicationWarning --> CommunicationOffline : lastUpdate age > 240,000 ms
```

```
    CommunicationOffline --> CommunicationWarning : telemetry snapshot received
```

```
    CommunicationOffline --> CommunicationOffline : tick / maintain last values
```

```
    Shutdown --> [*]
```

```
...
```

```
**States:**
```

State	Description	
-------	-------------	--

----- -----	
Startup	System power-on. Hardware peripherals initializing.
Initializing	Restoring persisted device state, plants, LED, irrigation, history.
WaitingForTelemetry	First sensor snapshot not yet received. Dashboard shows "No Data".
Operating	Normal operation. Telemetry being received and processed.
CommunicationWarning	Telemetry delayed > 120 s. Warning indicator shown.
CommunicationOffline	Telemetry absent > 240 s. Offline indicator shown.
Shutdown	System power-off sequence.

4. Startup and Initialization

```mermaid

stateDiagram-v2

[\*] --> HardwareBoot

HardwareBoot --> LoadStoredDevices : hardware ready

LoadStoredDevices --> LoadStoredNotifications : devices loaded

LoadStoredNotifications --> LoadTelemetryHistory : notifications loaded

LoadTelemetryHistory --> LoadProfileAndSettings : history loaded

LoadProfileAndSettings --> SeedDefaultDevice : [no devices found] / create default device

LoadProfileAndSettings --> SelectActiveDevice : [devices exist]

SeedDefaultDevice --> SelectActiveDevice : default device created

SelectActiveDevice --> StartTelemetryProvider : active device set

StartTelemetryProvider --> RenderInitialScreen : provider started

RenderInitialScreen --> [\*] : initial Home screen rendered

```

****Initialization sequence in firmware:****

1. ****HardwareBoot**** — Initialize display, peripherals, storage subsystem.
2. ****LoadStoredDevices**** — Read persisted device array from storage (`gl\_devices\_v1`). Restore: device metadata, plant config, LED config, irrigation config, tank fill level.
3. ****LoadStoredNotifications**** — Read notification log from storage (`gl\_notifications\_v1`).
4. ****LoadTelemetryHistory**** — Read rolling sample arrays from storage (`gl\_history\_v1`).
5. ****LoadProfileAndSettings**** — Read operator profile and app settings.
6. ****SeedDefaultDevice**** — If no devices exist, create one default device with factory defaults. See Document 1, Section 2.3 for default values.
7. ****SelectActiveDevice**** — Set `activeDeviceId` to the first device or the previously active device ID.
8. ****StartTelemetryProvider**** — Begin telemetry provider. Deliver first snapshot immediately.
9. ****RenderInitialScreen**** — Display Home screen. Sensor values show last persisted values or defaults.

****Important:**** Sensor values are ****transient**** and are ****not persisted**** between boots. After boot, sensor tiles will show the default initial values until the first telemetry snapshot is received from the hardware. The firmware must deliver a snapshot as soon as sensor readings are available.

5. Telemetry State Machine

```mermaid

stateDiagram-v2

[\*] --> Idle

Idle --> Sampling : telemetry tick fires (interval: 5,000 / 15,000 / 30,000 ms)

Sampling --> ApplySnapshot : sensor values read

Sampling --> SnapshotFailed : sensor read error

ApplySnapshot --> UpdateLastTimestamp : merge into sensors object / lastUpdate = now

UpdateLastTimestamp --> SyncConnectionMetadata : compute connectionStatus from age

SyncConnectionMetadata --> RecordHistorySample : [age since last sample > 10,000 ms]  
 SyncConnectionMetadata --> SkipHistorySample : [age since last sample ≤ 10,000 ms]  
 RecordHistorySample --> TriggerScreenRefresh  
 SkipHistorySample --> TriggerScreenRefresh  
 TriggerScreenRefresh --> GenerateNotifications : refresh Home/Insights screen  
 GenerateNotifications --> PersistIfDue : [time since last generation > 8,000 ms] / run alert engine  
 GenerateNotifications --> SkipNotifGen : [throttled]  
 PersistIfDue --> Idle : [every 4 ticks] / persist device and history  
 SkipNotifGen --> Idle  
 SnapshotFailed --> Idle : log error, no state change  
 ...

#### **\*\*Events:\*\***

| Event                | Source                        |
|----------------------|-------------------------------|
| telemetry tick fires | Hardware timer (periodic)     |
| sensor values read   | Sensor hardware / BLE / Cloud |
| sensor read error    | Hardware fault                |

#### **\*\*Guards:\*\***

| Guard                                 | Condition                           |
|---------------------------------------|-------------------------------------|
| age since last sample > 10,000 ms     | Throttle: one sample per 10 seconds |
| time since last generation > 8,000 ms | Throttle: notification generation   |
| every 4 ticks                         | Persistence: ~60 s at normal speed  |

#### **\*\*Outputs:\*\***

| Action | Description |
|--------|-------------|
|--------|-------------|

```

|-----|-----|
| merge into sensors | Assign ph, waterTemp, airTemp, humidity, cwbeLevel, cwbeStatus,
pumpActive |
lastUpdate = now	Set timestamp for staleness tracking
compute connectionStatus	online / warning / offline (see Section 15)
record sample	Append sample to rolling history buffer
refresh Home screen	Re-render dashboard if Home or Insights is visible
run alert engine	Evaluate all notification conditions
persist device	Write device state to storage

```

---

## ## 6. Sensor Validation State Machine

This state machine describes the per-sensor validation that runs for each field in the telemetry snapshot.

```
```mermaid
```

```
stateDiagram-v2
```

```
    [*] --> CheckPresence
```

```
    CheckPresence --> CheckRange : [value is numeric]
```

```
    CheckPresence --> MissingData : [value is null / NaN / undefined]
```

```
    CheckRange --> Valid : [value within physical plausibility bounds]
```

```
    CheckRange --> InvalidData : [value outside plausibility bounds]
```

```
    Valid --> ComputeInsight : apply getSensorInsight(key, value, stage)
```

```
    ComputeInsight --> StatusOK : [insight.status = 'ok']
```

```
    ComputeInsight --> StatusLow : [insight.status = 'low']
```

```
    ComputeInsight --> StatusWarn : [insight.status = 'warn']
```

StatusOK --> [*] : display value, no alert
StatusLow --> [*] : display value, informational indicator
StatusWarn --> [*] : display value, warning indicator, alert generated

MissingData --> [*] : display '-', suppress health rules
InvalidData --> [*] : display '-', suppress health rules, log fault

...

****Physical plausibility bounds per sensor:****

Sensor	Min	Max	
-----	-----	-----	
pH	0.0	14.0	
waterTemp	0 °C	50 °C	
airTemp	0 °C	50 °C	
humidity	0 %	100 %	
cwbeLevel	0 %	100 %	

Values outside these bounds are treated as `InvalidData`.

****Key rule:****

`getSensorInsight()` is only called when the value is numeric and within plausibility bounds. The insight result determines whether an alert will fire during the next notification generation cycle.

7. Health Evaluation State Machine

```mermaid

stateDiagram-v2

[\*] --> WaitingForTelemetry : [lastUpdate = null]



WaitingForTelemetry --> EvaluatingRules : [lastUpdate ≠ null] / first snapshot received

EvaluatingRules --> CollectFiredRules : evaluate all AGRONOMIST\_RULES

note right of CollectFiredRules

For each rule:

if rule.condition(dev) === true → add to firedRules

if dev.sensors.lastUpdate = null → condition returns false

end note

CollectFiredRules --> ComputePenaltyScore : sort firedRules by priority ascending

ComputePenaltyScore --> DetermineStatusFromScore : score = 100 – sum(penalties)

DetermineStatusFromScore --> DetermineStatusFromRules

DetermineStatusFromRules --> MergeStatus : final = max\_rank(scoreStatus, ruleStatus)

MergeStatus --> StatusExcellent : [no rules fired]

MergeStatus --> StatusGood : [only info rules]

MergeStatus --> StatusAttention : [any warning rules; no critical]

MergeStatus --> StatusCritical : [any critical rules]

StatusExcellent --> OutputReport

StatusGood --> OutputReport

StatusAttention --> OutputReport

StatusCritical --> OutputReport

OutputReport --> [\*] : { status, score, firedRules, confidence, trends, forecasts }

EvaluatingRules --> EvaluatingRules : on each telemetry tick

...

**\*\*Status rank for merge operation:\*\***

| Status    | Rank  |
|-----------|-------|
| -----     | ----- |
| excellent | 0     |
| good      | 1     |
| attention | 2     |
| critical  | 3     |

Final status = status with the highest rank from either scoring method.

**\*\*Score-to-status mapping:\*\***

| Score     | statusFromScore |
|-----------|-----------------|
| -----     | -----           |
| $\geq 85$ | excellent       |
| 65 – 84   | good            |
| 40 – 64   | attention       |
| 0 – 39    | critical        |

**\*\*Rule-to-status mapping:\*\***

| Highest severity in firedRules | statusFromRules |
|--------------------------------|-----------------|
| -----                          | -----           |
| critical                       | critical        |
| warning                        | attention       |
| info                           | good            |
| (none)                         | excellent       |

---

## ## 8. Alert Generation State Machine

```mermaid

stateDiagram-v2

[*] --> ThrottleCheck

ThrottleCheck --> Suppressed : [now - lastRun < 8,000 ms AND NOT forced]

ThrottleCheck --> BuildDesiredSet : [not throttled OR forced]

Suppressed --> [*]

BuildDesiredSet --> EvaluateReservoir : check cwbeLevel, daysRemaining

EvaluateReservoir --> EvaluateTelemetry : check lastUpdate age

EvaluateTelemetry --> EvaluatePH : check pH insight

EvaluatePH --> EvaluateWaterTemp : check waterTemp insight

EvaluateWaterTemp --> EvaluateAirTemp : check airTemp insight

EvaluateAirTemp --> EvaluateHumidity : check humidity insight

EvaluateHumidity --> EvaluateAlignment : check LED/irrigation alignment

EvaluateAlignment --> MergeWithExisting

MergeWithExisting --> PreserveOngoing : keep alerts whose dedupKey still in desiredSet

PreserveOngoing --> RemoveCleared : remove alerts whose dedupKey no longer in desiredSet

RemoveCleared --> AddNew : create new alert entries for newly fired conditions

AddNew --> TrimToLimit : [total count > 50] / drop oldest

TrimToLimit --> Persist : write to storage

Persist --> UpdateBadge : update bell badge count

UpdateBadge --> [*] : set lastRun = now

```

**\*\*Alert deduplication logic:\*\***

Each state-based alert carries a `dedupKey` (e.g., `reservoir-crit-dev-001`). On each generation run:

1. The desired set is computed fresh from current sensor values.
2. Existing alerts with dedupKeys that are still in the desired set are **\*\*preserved\*\*** (their `createdAt` and `read` status are retained). Their text is updated if the message has changed.
3. Existing alerts with dedupKeys that are **\*\*no longer\*\*** in the desired set are **\*\*removed\*\***.
4. Alerts in the desired set that are **\*\*not\*\*** in the existing set are **\*\*created\*\*** as new entries.
5. Event alerts (no dedupKey) are **\*\*never removed\*\*** by this process.

**\*\*Alert severity evaluation chain:\*\***

Condition Evaluated	Alert Key Pattern	Severity
-----	-----	-----
cwbeLevel ≤ 5 %	reservoir-crit-{did}	critical
daysRemaining < 1	refill-today-{did}	critical
daysRemaining ≤ 3	refill-soon-{did}	warning
lastUpdate age > 120,000 ms	telemetry-stale-{did}	critical
pH insight = 'warn'	ph-warn-{did}	warning
pH insight = 'low'	ph-low-{did}	warning
waterTemp insight = 'warn'	wt-warn-{did}	warning
airTemp insight = 'warn'	at-warn-{did}	warning
airTemp insight = 'low'	at-low-{did}	warning
humidity insight = 'warn'	rh-warn-{did}	warning
humidity insight = 'low'	rh-low-{did}	warning
irrigation/LED misalignment	irr-align-{did}	warning

---

**## 9. Recommendation Generation State Machine**

```mermaid

stateDiagram-v2

[*] --> CheckTelemetryAvailable

CheckTelemetryAvailable --> TelemetryOffline : [lastUpdate = null OR age > 300,000 ms]

CheckTelemetryAvailable --> EvaluateRules : [telemetry available]

TelemetryOffline --> [*] : output = { telemetryOffline: true, firedRules: [] }

EvaluateRules --> RulesPassed : for each rule in AGRONOMIST_RULES

RulesPassed --> SortByPriority : firedRules sorted ascending by priority number

SortByPriority --> ComputeHealth : _computeHealthFromRules(firedRules)

ComputeHealth --> ComputeTrends : analyze history samples

ComputeTrends --> ComputeConfidence : assess history coverage

ComputeConfidence --> ResolvePlantScope : collect all plant profiles from savedPlants

ResolvePlantScope --> ComputeCompatibility : cross-species range intersection

ComputeCompatibility --> ComputeDominant : identify dominant crop by site count

ComputeDominant --> ComputeLimiting : identify limiting crop by compatibility score

ComputeLimiting --> ComputeStress : compute per-species sensor deviation

ComputeStress --> AssembleReport

AssembleReport --> [*] : return AgronomistReport {

generatedAt, deviceId, confidence, health,

firedRules, historicalInsights, forecasts,

context, plantProfile, plantScope, cropCompatibility,

dominantCrop, limitingCropAnalysis, cropStress,

telemetryOffline, telemetryLastUpdate

}

```

**\*\*Trend computation (per parameter):\*\***

```
...
```

```
trendWindow = last 10 samples
```

```
if sampleCount < 3 → 'unknown'
```

```
split window into first half and second half
```

```
meanFirst = average of first half values
```

```
meanSecond = average of second half values
```

```
delta = meanSecond - meanFirst
```

```
if delta > threshold → 'rising'
```

```
if delta < -threshold → 'falling'
```

```
else → 'stable'
```

```
...
```

Thresholds: pH=0.20, waterTemp=0.50°C, airTemp=0.50°C, humidity=3.00%, cwbeLevel=3.00%

**\*\*Confidence computation:\*\***

```
...
```

```
if sampleCount < 3: { score: 0.00, label: 'none' }
```

```
coverageHours = (samples[-1].t - samples[0].t) / 3,600,000
```

```
if coverageHours >= 24: { score: 1.00, label: 'extended' }
```

```
if coverageHours >= 6: { score: 0.80, label: 'limited' }
```

```
if coverageHours >= 1: { score: 0.55, label: 'partial' }
```

```
else: { score: 0.30, label: 'live' }
```

```
...
```

```

```

**## 10. Dashboard Refresh State Machine**

```
```mermaid
```

```
stateDiagram-v2
```

```
    [*] --> Idle
```

```
    Idle --> RenderingHome : [active screen = 'home'] / telemetry tick or action
```

```
    Idle --> RenderingInsights : [active screen = 'cwbe'] / telemetry tick or action
```

```
    Idle --> RenderingOther : [active screen = other] / action only
```

```
    RenderingHome --> BuildHomeScreen : read active device state
```

```
    BuildHomeScreen --> RenderCWBEHero : compute lvl, cwbeStatus, forecast
```

```
    RenderCWBEHero --> RenderSensorGrid : pH, waterTemp, airTemp, humidity tiles
```

```
    RenderSensorGrid --> RenderPumpCard : pumpActive state
```

```
    RenderPumpCard --> RenderQuickNav : navigation tiles
```

```
    RenderQuickNav --> CommitToDisplay : assemble complete screen
```

```
    CommitToDisplay --> Idle
```

```
    RenderingInsights --> BuildInsightsScreen : read device state + history
```

```
    BuildInsightsScreen --> RunAgronomistReport : getAgronomistReport(dev)
```

```
    RunAgronomistReport --> RenderAgroCard : assemble observation blocks
```

```
    RenderAgroCard --> CommitToDisplay
```

```
    CommitToDisplay --> Idle
```

```
    RenderingOther --> BuildOtherScreen : read relevant device state
```

```
    BuildOtherScreen --> CommitToDisplay
```

```
```
```

### **\*\*Key rule:\*\***

Telemetry ticks only trigger a re-render if the active screen is 'home' or 'cwbe' (Insights). Other screens (LED editor, irrigation editor, plants picker) are not refreshed by telemetry ticks — only by explicit operator actions.

This prevents screen disruption while the operator is editing configuration.

---

## ## 11. Screen Navigation State Machine

```mermaid

stateDiagram-v2

[*] --> Home

Home --> Plants : navigate('plants') / discard draft

Home --> LED : navigate('led') / discard draft

Home --> Irrigation : navigate('irrigation') / discard draft

Home --> Insights : navigate('cwbe') / discard draft

Home --> Profile : navigate('profile')

Home --> Notifications : bell tap / open notification sheet

Home --> DeviceSelector : device name tap / open device selector sheet

Plants --> Home : navigate('home') / discard plant draft

Plants --> PlantPicker : pick plant action

PlantPicker --> Plants : confirm or cancel

LED --> Home : navigate away / [editing=false] discard LED draft

LED --> LEDEditor : edit button / ledEditing = true

LEDEditor --> LED : save OR cancel / ledEditing = false

Irrigation --> Home : navigate away / discard irrigation draft

Irrigation --> IrrigationEditor : edit button / irrigationEditing = true

IrrigationEditor --> Irrigation : save OR cancel / irrigationEditing = false

Insights --> Home : navigate('home')

Profile --> Home : back button

Notifications --> Home : backdrop tap or swipe down

DeviceSelector --> Home : device selected or dismissed

...

****discard draft behavior on navigation:****

Each screen maintains an unsaved draft. When the operator navigates away, the draft is discarded and the saved configuration is restored.

| | | |
|------------|---|--|
| Screen | Draft Fields Discarded | |
| ----- | ----- | |
| Plants | dev.plants restored from dev.savedPlants; picker closed | |
| LED | dev.led restored from dev.savedLed; ledEditing = false | |
| Irrigation | dev.irrigation restored from dev.savedIrrigation; irrigationEditing = false; pump timer cleared | |

Navigation actions always reset the scroll position of the destination screen to the top.

12. Touch Interaction State Machine

```mermaid

stateDiagram-v2

[\*] --> WaitingForTouch

WaitingForTouch --> TouchDetected : touch down event

TouchDetected --> IdentifyTarget : determine data-action attribute

IdentifyTarget --> DispatchAction : [valid action found] / call Actions[action](data)

IdentifyTarget --> Ignore : [no action found]

DispatchAction --> MutateState : action modifies appState

MutateState --> TriggerRender : re-render affected screen

TriggerRender --> WaitingForTouch

Ignore --> WaitingForTouch

...

### **\*\*Action dispatch mechanism:\*\***

All interactive elements carry a `data-action` attribute that names the handler in the `Actions` object, plus optional `data-\*` attributes that provide parameters.

Examples:

| data-action          | Parameters  | Handler Behavior                           |  |
|----------------------|-------------|--------------------------------------------|--|
| -----                | -----       | -----                                      |  |
| navigate             | data-screen | Navigate to named screen                   |  |
| togglePump           | (none)      | Invert pumpActive; show toast              |  |
| adjustTank           | data-delta  | Adjust tankLiters by delta or preset       |  |
| saveLedConfig        | (none)      | Commit led draft to savedLed               |  |
| saveIrrigationConfig | (none)      | Commit irrigation draft to savedIrrigation |  |
| openDeviceSelector   | (none)      | Open device selector overlay               |  |

### **\*\*Touch active feedback:\*\***

Interactive elements apply an active opacity (typically 0.55–0.70) on press. Interactive cards apply a scale transform (0.965) on press. This feedback must be implemented in firmware UI rendering.

---

## **## 13. LED Configuration State Machine**

```
```mermaid
```

```
stateDiagram-v2
```

```
    [*] --> LedSummaryView
```

```
    LedSummaryView --> LedEditorView : tap "Edit LED" / ledEditing = true
```

```
    LedEditorView --> SelectPreset : tap preset button
```

```
    LedEditorView --> EditOnTime : tap ON time field
```

```
    LedEditorView --> EditOffTime : tap OFF time field
```

```
    LedEditorView --> AdjustIntensity : slide intensity slider
```

```
    SelectPreset --> LedEditorView : preset applied / fill on, off, intensity from LED_PRESETS[preset]
```

```
    note right of SelectPreset
```

```
        Selecting Seedling/Standard/Bloom/Off
```

```
        overwrites on, off, intensity with preset values.
```

```
        Manually changing any field sets preset = 'custom'.
```

```
    end note
```

```
    EditOnTime --> LedEditorView : time confirmed
```

```
    EditOffTime --> LedEditorView : time confirmed
```

```
    AdjustIntensity --> LedEditorView : slider released / update intensity, set preset = 'custom' if not  
    matching any preset
```

```
    LedEditorView --> CommitLed : tap "Save LED" / [valid times]
```

```
    LedEditorView --> DiscardLed : tap "Cancel"
```

```
    CommitLed --> LedSummaryView : dev.savedLed = { ...dev.led }; ledEditing = false; persist; show  
    toast
```

```
    DiscardLed --> LedSummaryView : dev.led = { ...dev.savedLed }; ledEditing = false
```

```
    LedSummaryView --> LedSummaryView : telemetry tick / re-render status (active/scheduled/off)
```

```
```
```

### **\*\*Preset auto-detection on save:\*\***

...

detectLedPreset(led):

for each preset in LED\_PRESETS:

if led.on = preset.on AND led.off = preset.off AND led.intensity = preset.intensity:

return preset\_key

return 'custom'

...

This is applied when displaying the preset badge after save. It is not a gate on saving.

### **\*\*LED status re-evaluation:\*\***

After every save, the LED status (active / scheduled / off) is recomputed from the current system clock against the saved ON/OFF times. Status can change without any operator action (transitions at ON time and OFF time each day).

---

## **## 14. Irrigation Configuration State Machine**

```mermaid

stateDiagram-v2

[*] --> IrrigationSummaryView

IrrigationSummaryView --> IrrigationEditorView : tap "Edit irrigation" / irrigationEditing = true

IrrigationEditorView --> SelectSubstrate : tap substrate card

IrrigationEditorView --> SelectStage : tap stage selector

IrrigationEditorView --> AdjustCycleOn : tap +/- stepper for ON time

IrrigationEditorView --> AdjustCyclePause : tap +/- stepper for PAUSE time

IrrigationEditorView --> ApplyRecommended : tap "Apply recommended"

SelectSubstrate --> IrrigationEditorView : substrate updated / recompute recommended cycle

SelectStage --> IrrigationEditorView : stage updated / recompute recommended cycle

AdjustCycleOn --> IrrigationEditorView : cycleOn updated

AdjustCyclePause --> IrrigationEditorView : cyclePause updated / check substrate warnings

ApplyRecommended --> IrrigationEditorView : fill cycleOn, cyclePause from
getRecommendedCycle(stage, substrate)

IrrigationEditorView --> CommitIrrigation : tap "Save" / [valid config]

IrrigationEditorView --> DiscardIrrigation : tap "Cancel"

CommitIrrigation --> IrrigationSummaryView : savedIrrigation updated; irrigationSaved = true;
persist; show toast; generate notifications (forced)

DiscardIrrigation --> IrrigationSummaryView : restore from savedIrrigation; irrigationEditing = false

...

****getRecommendedCycle computation:****

...

base = IRRIGATION_MATRIX[stage][substrate]

tempCategory = getTempCategory(dev.sensors.airTemp)

cropKey = getCropKey(dominantPlantId)

tempMod = TEMP_OFF_MODIFIERS[tempCategory] // 1.15 / 1.00 / 0.82 / 0.67

cropDiv = CROP_OFF_DIVISORS[cropKey] // 1.00 / 0.95 / 0.80 / 0.73 / 0.68 / 1.20

recommendedON = base.on

recommendedPAUSE = min(360, max(15, round(base.off × tempMod / cropDiv)))

...

****cyclePause clamps:**** minimum 15 min, maximum 360 min (6 hours).

****cycles per day formula:****

...

period = cycleOn + cyclePause

if nightBreak AND windowStart AND windowEnd:

 windowMins = (windowEnd > windowStart) ? offMins-onMins : 1440-onMins+offMins

 cyclesPerDay = max(0, floor(windowMins / period))

else:

 cyclesPerDay = floor(1440 / period)

...

****runtime per day:****

...

runtimePerDay = cyclesPerDay × cycleOn (minutes)

...

15. Communication Failure and Recovery State Machine

``mermaid

stateDiagram-v2

 [*] --> Online

Online --> Online : telemetry tick received within 120,000 ms / update lastUpdate

Online --> Warning : lastUpdate age > 120,000 ms / connectionStatus = 'warning'

Warning --> Online : telemetry tick received / connectionStatus = 'online' / clear stale alert

Warning --> Offline : lastUpdate age > 240,000 ms / connectionStatus = 'offline'

Offline --> Warning : telemetry tick received / connectionStatus = 'warning'
Offline --> Offline : time passes / maintain last sensor values / stale alert active

Online --> AgronomistOnline : lastUpdate age $\leq$ 300,000 ms
AgronomistOnline --> AgronomistOffline : lastUpdate age > 300,000 ms

AgronomistOffline --> AgronomistOnline : telemetry tick received
AgronomistOffline --> AgronomistOffline : crop analysis suspended

...

****Thresholds:****

| Threshold | Value | Effect |
|--------------------|------------|---|
| ----- | ----- | ----- |
| Stale | 120,000 ms | connectionStatus $\rightarrow$ 'warning'; stale alert generated |
| Offline | 240,000 ms | connectionStatus $\rightarrow$ 'offline' |
| Agronomist offline | 300,000 ms | Plant analysis suspended |

****Sensor display during communication failure:****

Last received sensor values remain displayed (not cleared). Only the live indicator (animated dot) distinguishes live data from stale display.

****Alert behavior:****

The `telemetry-stale-{did}` alert is generated when `lastUpdate age > 120,000 ms`. It is automatically removed from the active alert set when a new snapshot arrives and the next notification generation run executes.

****Recovery:****

Any valid telemetry snapshot received resets `lastUpdate` to the current timestamp. All status thresholds are then recomputed from the new age.

16. Pump Manual Control State Machine

```mermaid

stateDiagram-v2

[\*] --> PumpIdle : pumpActive = false

PumpIdle --> PumpManualRunning : operator tap togglePump / pumpActive = true; send hardware command; show toast "Pump activated"

PumpManualRunning --> PumpIdle : operator tap togglePump / pumpActive = false; send hardware command; show toast "Pump stopped"

PumpIdle --> PumpIdle : telemetry tick [pumpActive = false from hardware]

PumpManualRunning --> PumpManualRunning : telemetry tick [pumpActive = true from hardware]

PumpManualRunning --> PumpIdle : telemetry update [pumpActive = false] / hardware stopped pump

...

**\*\*States:\*\***

| State             | sensors.pumpActive | Display Label      |
|-------------------|--------------------|--------------------|
| -----             | -----              | -----              |
| PumpIdle          | false              | Idle · on schedule |
| PumpManualRunning | true               | Running · manual   |

**\*\*Hardware command:\*\***

When the operator toggles the pump, a hardware command is dispatched immediately. The telemetry snapshot from hardware subsequently confirms the actual state.

---

## ## 17. Tank Adjustment State Machine



```
```mermaid
```

```
stateDiagram-v2
```

```
    [*] --> TankAtLevel
```

```
TankAtLevel --> TankAdjusting : operator taps adjustment button
```

```
TankAdjusting --> ComputeTarget : determine target from delta or preset
```

```
note right of ComputeTarget
```

```
    delta = '-0.5' → target = current - 0.5
```

```
    delta = '+0.5' → target = current + 0.5
```

```
    delta = '+1.0' → target = current + 1.0
```

```
    delta = 'nominal' → target = 6.0 L
```

```
    delta = 'max' → target = 6.5 L
```

```
end note
```

```
ComputeTarget --> ClampTarget : target = clamp(target, 0, 6.5)
```

```
ClampTarget --> UpdateTankState : tankLiters = clamped target
```

```
UpdateTankState --> RecomputeCWBE : lvl = round(tankLiters / 6.5 × 100)
```

```
RecomputeCWBE --> UpdateCWBEStatus : status = lvl < 15 ? 'critical' : lvl < 30 ? 'low' : 'ok'
```

```
UpdateCWBEStatus --> PersistDevice
```

```
PersistDevice --> LogEventNotification : addEventNotification('refill', 'info', ...)
```

```
LogEventNotification --> Render : re-render Home screen
```

```
Render --> TankAtLevel
```

```
...
```

```
---
```

18. Notification Center State Machine

```
```mermaid
```

```
stateDiagram-v2
```

```
 [*] --> Closed
```

```
 Closed --> Opening : bell tap
```

```
 Opening --> Open : sheet animation complete
```

```
 Open --> MarkingRead : tap notification card
```

```
 MarkingRead --> Open : notification.read = true; persist; update badge
```

```
 Open --> Clearing : tap "Clear All"
```

```
 Clearing --> Open : notifications = []; persist; badge = 0
```

```
 Open --> Closing : tap backdrop or swipe down
```

```
 Closing --> Closed : sheet animation complete
```

```
 Open --> Open : new alert arrives / re-render list
```

```
...
```

```
Badge count computation:
```

```
...
```

```
unreadCount = notifications.filter(n => !n.read).length
```

```
if unreadCount = 0: hide badge
```

```
if unreadCount > 0: display badge with count
```

```
...
```

Badge is always visible when `unreadCount > 0`, regardless of which screen is active.

```

```

## ## 19. Future Integration Points

### ### 19.1 CWBE Integration Point

The CWBE (Crop Water Balance Efficiency) module will be integrated as an extension to the Health Evaluation State Machine without requiring structural changes.

```mermaid

stateDiagram-v2

[*] --> EvaluateRules

EvaluateRules --> EvaluateSensorRules : existing pipeline (rules 10-51)

EvaluateSensorRules --> EvaluateCWBERules : [FUTURE] CWBE rules (priority 60-69)

EvaluateCWBERules --> ComputeHealth

ComputeHealth --> [*]

```

#### **\*\*Reserved integration state in Health Evaluation:\*\***

After rule priority group 51 (reservoir rules), a CWBE rule group at priorities 60–69 will be added. No existing rule priorities need to change. The health score computation and status derivation remain identical.

**\*\*Reserved telemetry field:\*\*** `sensors.cwbeLevel` (currently used as reservoir percentage) will be supplemented or replaced by the CWBE efficiency score. The history sample already records `cwbeLevel`, providing continuity.

### ### 19.2 AI Assistant Integration Point

The current Agronomist chat processes questions through a deterministic rule-based router. The AI integration point replaces the response generation step without changing the input/output interface.

```
```mermaid
```

```
stateDiagram-v2
```

```
    [*] --> ReceiveQuestion
```

```
    ReceiveQuestion --> BuildContext : buildAgronomistContext(dev)
```

```
    BuildContext --> RouteIntent : classify question intent
```

```
    RouteIntent --> RuleBasedResponse : [V1: rule engine]
```

```
    RouteIntent --> AIBackendRequest : [FUTURE: LLM API call]
```

```
    RuleBasedResponse --> DisplayResponse
```

```
    AIBackendRequest --> DisplayResponse
```

```
    DisplayResponse --> [*]
```

```
```
```

The `buildAgronomistContext(dev)` function and the chat history/display infrastructure remain unchanged. Only the response generation function is replaced.

---

## ## 20. State Summary Tables

### ### 20.1 All Named System States

| State Machine | State               | Description                 |  |
|---------------|---------------------|-----------------------------|--|
| -----         | -----               | -----                       |  |
| System        | Startup             | Hardware power-on           |  |
| System        | Initializing        | Restoring stored state      |  |
| System        | WaitingForTelemetry | No sensor data yet received |  |

|            |                      |                                             |  |
|------------|----------------------|---------------------------------------------|--|
| System     | Operating            | Normal telemetry-driven operation           |  |
| System     | CommunicationWarning | Telemetry delayed >120 s                    |  |
| System     | CommunicationOffline | Telemetry absent >240 s                     |  |
| System     | Shutdown             | Power-off sequence                          |  |
| Telemetry  | Idle                 | Between ticks                               |  |
| Telemetry  | Sampling             | Sensor read in progress                     |  |
| Telemetry  | ApplySnapshot        | Merging values into device state            |  |
| Telemetry  | RecordHistorySample  | Appending sample to rolling buffer          |  |
| Telemetry  | SnapshotFailed       | Sensor read error                           |  |
| Health     | WaitingForTelemetry  | lastUpdate = null; no evaluation            |  |
| Health     | EvaluatingRules      | Rule conditions being evaluated             |  |
| Health     | StatusExcellent      | Score $\geq 85$ , no warning/critical rules |  |
| Health     | StatusGood           | Score $\geq 65$ , only info rules           |  |
| Health     | StatusAttention      | Score $\geq 40$ , warning rules active      |  |
| Health     | StatusCritical       | Critical rules active                       |  |
| Alert      | ThrottleCheck        | Checking generation interval                |  |
| Alert      | Suppressed           | Generation skipped (throttled)              |  |
| Alert      | BuildDesiredSet      | Evaluating all alert conditions             |  |
| Alert      | MergeWithExisting    | Reconciling desired vs. current alerts      |  |
| Navigation | Home                 | Home screen active                          |  |
| Navigation | Plants               | Plant configuration screen                  |  |
| Navigation | LED                  | LED summary screen                          |  |
| Navigation | LEDEditor            | LED configuration editor open               |  |
| Navigation | Irrigation           | Irrigation summary screen                   |  |
| Navigation | IrrigationEditor     | Irrigation configuration editor open        |  |
| Navigation | Insights             | Water stats / Agronomist screen             |  |
| Navigation | Profile              | Operator profile screen                     |  |
| Navigation | Notifications        | Notification center sheet open              |  |
| Navigation | DeviceSelector       | Device picker sheet open                    |  |
| Pump       | PumpIdle             | Pump not running                            |  |
| Pump       | PumpManualRunning    | Pump running via manual command             |  |

|               |                   |                                        |  |
|---------------|-------------------|----------------------------------------|--|
| Communication | Online            | Telemetry fresh (age ≤ 120 s)          |  |
| Communication | Warning           | Telemetry stale (120–240 s)            |  |
| Communication | Offline           | Telemetry absent (> 240 s)             |  |
| Communication | AgronomistOnline  | Plant analysis active (age ≤ 300 s)    |  |
| Communication | AgronomistOffline | Plant analysis suspended (age > 300 s) |  |

### ### 20.2 All State Transitions with Guards

| From State<br>Output                           | Event<br>         | Guard         | To State             |                    |
|------------------------------------------------|-------------------|---------------|----------------------|--------------------|
| -----                                          | -----             | -----         | -----                | -----              |
| Startup<br>load                                | power on          | —             | Initializing         | begin stored state |
| Initializing<br>active device                  | state loaded      | devices exist | WaitingForTelemetry  | select             |
| Initializing<br>device                         | state loaded      | no devices    | Initializing         | create default     |
| WaitingForTelemetry<br>snapshot, render        | snapshot received | —             | Operating            | apply              |
| Operating<br>render                            | tick              | age ≤ 120 s   | Operating            | apply snapshot,    |
| Operating<br>warning indicator, stale alert    | age check         | age > 120 s   | CommunicationWarning |                    |
| CommunicationWarning<br>stale alert            | snapshot received | —             | Operating            | clear              |
| CommunicationWarning<br>offline indicator      | age check         | age > 240 s   | CommunicationOffline |                    |
| CommunicationOffline<br>begin recovery         | snapshot received | —             | CommunicationWarning |                    |
| LED Summary<br>true                            | tap edit          | —             | LED Editor           | ledEditing =       |
| LED Editor<br>savedLed, toast, generate alerts | tap save          | —             | LED Summary          | commit             |
| LED Editor<br>draft                            | tap cancel        | —             | LED Summary          | discard            |

|                                       |                          |                           |                           |                    |
|---------------------------------------|--------------------------|---------------------------|---------------------------|--------------------|
| LED Editor                            | tap preset               | preset ≠ current          | LED Editor                | fill               |
| on/off/intensity from preset          |                          |                           |                           |                    |
| Irrigation Summary                    | tap edit                 | —                         | Irrigation Editor         |                    |
| irrigationEditing = true              |                          |                           |                           |                    |
| Irrigation Editor                     | tap save                 | —                         | Irrigation Summary        | commit             |
| savedIrrigation, irrigationSaved=true |                          |                           |                           |                    |
| Irrigation Editor                     | tap cancel               | —                         | Irrigation Summary        | discard            |
| draft                                 |                          |                           |                           |                    |
| Pump Idle                             | tap toggle               | —                         | Pump Running              |                    |
| pumpActive=true, hw command, toast    |                          |                           |                           |                    |
| Pump Running                          | tap toggle               | —                         | Pump Idle                 |                    |
| pumpActive=false, hw command, toast   |                          |                           |                           |                    |
| Pump Running                          | telemetry update         | hardware pumpActive=false | Pump                      |                    |
| Idle                                  | reflect hardware state   |                           |                           |                    |
| Notification Center                   | tap backdrop             | —                         | Closed                    | sheet dismiss      |
| animation                             |                          |                           |                           |                    |
| Notification Center                   | tap entry                | —                         | Open                      | mark read,         |
| update badge                          |                          |                           |                           |                    |
| Notification Center                   | tap clear all            | —                         | Open                      | clear all, badge = |
| 0                                     |                          |                           |                           |                    |
| Health Excellent                      | any info rule fires      | —                         | Health Good               | update             |
| display                               |                          |                           |                           |                    |
| Health Good                           | any warning rule fires   | —                         | Health Attention          | update             |
| display                               |                          |                           |                           |                    |
| Health Any                            | any critical rule fires  | —                         | Health Critical           | update             |
| display, alert generated              |                          |                           |                           |                    |
| Health Critical                       | all critical rules clear | no warning rules          | Health Attention or lower |                    |
| update display                        |                          |                           |                           |                    |
| Health Critical                       | all critical rules clear | only info rules           | Health Good               | update             |
| display                               |                          |                           |                           |                    |
| Health Critical                       | all rules clear          | —                         | Health Excellent          | update             |
| display                               |                          |                           |                           |                    |

---

*\*End of Document — GL Towers Dashboard State Machine Specification v1.0\**

